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Localization algorithm of wireless sensor network based on matrix reconstruction

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ABSTRACT

In view of the situation that the sensor network is applied to the complex environment, the connectivity is low, the ranging information is limited, and the ranging data has noise, this paper propose the positioning algorithm based on the addition of the generalized Lagrange matrix. Based on the low-rank characteristics of the distance matrix between the nodes of the sensor network, the distance recovery problem under partial sampling information is modeled as the matrix completion problem under Gaussian noise. The unmeasured distance information can be inferred through matrix filling and the noise and anomaly caused by the external environment can be filtered at the same time, which reduces the influence of environmental noise on the positioning and improves the positioning stability and accuracy.

1. Introduction

In recent years, wireless sensor networks (WSNs) technology has developed rapidly, and it is widely used in military reconnaissance, intelligent traffic, environmental monitoring and other fields. Smart home system, automatic parking distance warning, forest fire monitoring and other typical applications of WSNs. In WSNs, the acquisition of node location information is crucial for various applications. Limited by node energy, deployment conditions and economic factors, WSNs generally only has a few anchor nodes acquire their own positions by loading GPS, etc., while the location information of other unknown nodes is calculated by the positioning algorithm. At present, there are many WSNs self-positioning solutions, such as convex programming algorithm, DV-HOP algorithm, MSVR algorithm, etc [1]. On the whole, the node location of sensor network can be divided into two categories: location algorithm based on distance measurement and location algorithm independent of distance measurement. The former can realize more accurate positioning, but the cost of calculation and communication is large, and need some hardware support. The latter has lower positioning accuracy, but less computing overhead, and is suitable for low power consumption and low cost applications [2].

In order to improve the positioning accuracy, the distance square matrix is divided into modeled as the low-rank distance square matrix and the sparse noise matrix, and propose the augmented Lagrange matrix filling algorithm to fill the unmeasured distance on the basis of the measured value and separate the noise data. Based on the filled distance matrix, MDS method is used to realize the accurate location of nodes. Compared with the existing positioning algorithm based on matrix filling, the method in this paper can separate noise and anomaly during the reconstruction of the distance square matrix and realize the accurate estimation of the distance, thus improving the accuracy of the positioning algorithm. Simulation experiment verifies the performance of the proposed algorithm.

In this paper, the main research content of wireless sensor network node location problem is how to use augmented Lagrangian matrix to fill and reconstruct the complete Euclidean distance matrix, and establish a complete mathematical model. The complex situations in the environment, is capable of measuring data is very limited, this method is easy to reconstruct the Euclidean distance matrix of high precision, in the case of high connectivity, control the measurement data of a certain number set can effectively reduce the distance of communication and computation cost, and then using MDS, and deploy a small amount of beacon node localization.

The paper is divided into six parts, and the content arrangement of each part is as follows: the first part discusses the research background and significance of wireless sensor network positioning algorithm, introduces the main work and innovation points of this paper, and summarizes the structure of the paper. The second part summarizes the relevant research progress at home and abroad; The third part briefly introduces the basic model of wireless sensor network location problem; The fourth part briefly introduces matrix filling theory and performance evaluation index; The fifth part, this paper based on the augmented Lagrangian matrix filling adaptive sensor network localization algorithm, using matrix filling infer most did not measure distance information, at the same time filter out noise processing to the collected

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data, through the adaptive method to adapt to different scenarios, as well as carry out the simulation experiment to the algorithm, proves the algorithm; The last part is summarizes and prospect.

2. Related work

In recent years, matrix filling theory [3] has brought a revolutionary breakthrough for the restoration of low-rank matrix with a lot of missing information. By using the sparsity of the singular value vector of the matrix, the low-rank matrix is recovered by sampling some elements of the matrix. With the deployment of wireless sensor network to more complex application scenarios, the connectivity of the network is greatly reduced, resulting in less and less distance information between nodes that can be measured, that is, only part of the measured data can be obtained.

In 2010, Cheng Feng et al. a foreign researcher, proposed for the first time to reconstruct Euclidean distance matrix with the rank of kernel norm minimization matrix for positioning [4]. Will be able to measure the Euclidean distance as sample collection, data modeling Euclidean distance square matrix, because low Euclidean distance square matrix rank matrix, will be able to measure the distance to the collection as a sampling matrix, through the special nuclear minimum norm to solve the above optimization problem, get complete Euclidean distance matrix of no noise, and then by multidimensional scaling (MDS) [5] to calculate all the unknown node location information. In the same year, Andrea Montarary et al. a foreign researcher, proposed the algorithm OPTSPACE POSITIONINGm [6] for positioning by matrix filling. Different from the kernel norm minimization method of Cheng Feng et al. OPTSPACE POSITIONINGm is based on the MDS-MAP algorithm, optimize the final coordinates, and make use of the gradient descent matrix filling algorithm OTSPACE optimize the coordinates, and obtain the corrected coordinates.

Domestic research started very late. In 2015, researchers such as Fu Xiao first published the regularization matrix-based filling on INFOCOM [7], which was the latest research results of positioning based on matrix filling. After 2015, Xiao Fu et al. published the positioning algorithm of wireless sensor network based on norm regularization matrix completion again on computer research and development in 2016 [8].

The positioning algorithm of wireless sensor network based on matrix filling is rarely studied. It is developed by researchers with the development of matrix filling theory and the current situation of limited data collected by wireless sensor network itself, which brings opportunities for researchers to carry out research.

3. Target positioning problem

This paper studies multi — target localization of wireless sensor network based on sound signal energy. The multi-sensor nodes collect the sound signals emitted by the sound source target in the measured area, collect the sound signal energy from multi-point sampling, carry out cooperative information transmission and processing, and realize the target positioning [2]. It is assumed that there are M audio sensor nodes and K sound source targets in the measured area. Among them, the sound source target signal is a point source isotropic signal, and the signal intensity of the measured point is inversely proportional to the distance between the target source and the measured point. It is assumed in this paper that the signals of K signal sources are linear superposition and there is no signal interference. Then i (i = 1, 2 , the measured value of M) sensor nodes at time n is:

$$x_i(n) = s_i(n) + v_i(n) \tag{1}$$

In the formula:

$$s_i(n) = \gamma_i \sum_{k=1}^{k} \frac{a_k(n - t_{ki})}{\|\rho(n - t_{ki}) - r_i\|} \tag{2}$$

In the formula γ_i is the gain of sensor node. $v_i(n)$ is the zero mean white Gaussian noise, $a_k(n - t_{ki})$ is the signal strength measured at 1 m from the k-th sound source target, t_{ki} is the time delay of sound signal transmission from the k sound source target to the i sound source target. ρ_k is the position vector of the k sound source target, r_i is the position vector of the i node. The signal energy is obtained by sampling the measured values over a period of time and averaging them, The energy of the sound signal received by node i can be expressed as:

$$y_i(t) = y_{si}(t) + \varepsilon_i(t) = g_i \sum_{k=1}^{k} \frac{s_k(t)}{d_{ik}^2(t)} + \varepsilon_i(t) \tag{3}$$

The most important component of wireless sensor network is sensor node. Generally speaking, it is a small embedded system with sensing capability and wireless communication capability [9]. More sensor nodes are usually arranged in the monitoring area to obtain more comprehensive data of the target area. The scalability of sensor network means: firstly, sensor nodes are deployed in a random way, so the distribution range is very broad; Secondly, the density of sensor nodes in the geographic area per unit area is very high [10].

Large-scale sensor networks can obtain more comprehensive information from different spatial perspectives, reduce the accuracy requirements of single node data, and expand coverage and reduce blind spots. In the layout process of wireless sensor network, the connection relation between nodes is not known in advance. In addition, during the operation of the network, some nodes will fail due to energy exhaustion or environmental impact, and some new nodes will be added to the network. This requires that nodes can be self-organizing and manage node configuration automatically. At the same time, when the number of nodes changes, the network can adapt to the dynamic change of the topology structure. At present, the most widely used Internet is an address-centered network. Users must have the IP address of resources in order to access the resources in the Internet. Unlike the Internet, sensor networks core is data. Nodes in the network adopt number to identify. When users use sensor network to query events, they only need to inform the network of events, that is, the event itself is taken as the query condition. This data — based query is more suitable for wireless sensor networks. The sensor node mechanism is simple and usually includes the following main modules: sensor module, processor module, wireless communication module and energy supply module [2]. This paper studies the target positioning problem. Assuming that the target position does not change with time, the variable in Eq. (3) is independent of time t. At the same time, it is assumed that the gain factor of multi-sensor nodes is the same and the intensity value of multi-target sound signal is the same and constant, that is, it does not change with the target k and time t.

The target location problem of wireless sensor network is to solve the target's spatial position according to the measured values of multiple sensor nodes. By modeling the positioning problem as a matrix reconstruction problem, the target location can be achieved by solving the target location matrix vector [11]. In various applications of wireless sensor networks, the location information of sensor nodes is very important. Data without location information is no value in applications such as defense, military, environmental monitoring, industrial control, and smart home. Due to the cost limitation of wireless sensor nodes, cannot directly use GPS to realize positioning. From the perspective of whether distance information is needed or not, existing positioning algorithms can be divided into distance-based positioning algorithms and distance-independent positioning algorithms. The distance independent positioning algorithm is simple to implement and fast to calculate, but its accuracy is low; The distance-based positioning algorithm has higher accuracy, but it needs distance measurement, which puts forward higher requirements on the hardware of nodes.

In wireless sensor network, the operation of sensor node to determine its spatial position by using a few nodes with known position and some measured values (distance or Angle) according to a certain mechanism is called node localization [2]. According to whether the sensor

node knows its own position and coordinate information, the sensor node is divided into anchor node and common node: Anchor node refers to a small number of nodes in the network that have identified their own position coordinates. Generally, they can obtain their position coordinate information by carrying GPS device or manual deployment in advance, which generally only accounts for a small proportion in the whole network;Unknown node refers to other nodes in the network that cannot directly obtain their own location information [9]. Positioning algorithm is the main content of this paper. Firstly introduced the basic principle of matrix filling theory, and then detailed introduce the target positioning method based on matrix reconstruction.

4. Matrix filling theory and performance evaluation index

In many practical problems, data is often presented in the form of a matrix to facilitate the understanding, modeling, processing and analysis of the data. However, these data are usually missing and face a high degree of incompleteness. The problem to be solved in matrix filling is how to obtain relatively accurate and well-structured overall data through processing and analysis [12]. Intuitively speaking, for a matrix, we can only sample a part of the elements of the matrix, while a large part of other elements cannot be obtained due to various reasons. Therefore, it is a problem to be solved that how accurately and reasonably fill in the vacant elements. Due to the high lack of information, it is possible to restore the data with some additional conditions. The feasibility of matrix filling requires the low rank of matrix as the precondition [13]. This is also a point of communication between the two, that is, the overall data information can be expressed in a simple form, and that is, the number of variables that determine the characteristics of the data is very few.

4.1. The definition of fill in the matrix

The standard matrix filling problem is usually described as:

$$\begin{aligned} &\min_{X\in R^{n\times m}} rank(X) \\ &s.t. P_{\Omega}(M) = P_{\Omega}(X) \end{aligned} \tag{4}$$

And Ω is the subscript set of elements, $P_{\Omega}(\cdot)$ is the orthogonal projection operator, which means when $i, j) \in \Omega$, M_{ij} is the sampling element. Since the rank of the matrix is a non-convex function, the problem is NP-hard, and the time required to solve the problem increases exponentially as the size of the matrix increases. Inspired by compression perception theory, Candes et al. [14]., the nuclear norm of the matrix are introduced to replace the matrix rank function, and Recht and others are theoretically proved that the function of matrix rank can be replaced by its convex hull nuclear norm [14], therefore, matrix filling problem is in to ensure good structural characteristics of low rank matrix under matrix of element height loss data completion.

4.2. Feasibility analysis of matrix filling

If M is a square $n * n$ matrix of rank r, then M contains a total of n^2 elements, but the singular value decomposition shows that it has $2n - r) * r$ degree of freedom. When the rank r is small, the degree of freedom is much smaller than the number elements. The ratio of degree of freedom to number of elements reflects the dimensionality reduction ratio of practical problems [15]. Since the dimensions of the problem are substantially lower than those on the surface, it is possible to consider obtaining the overall shape of the matrix through a small number of data samples.

Although the premise of low rank makes matrix filling possible, not all low-rank matrices can be filled with a small number of samples. In the matrix filling problem, the correlation between the span of singular vectors should also be considered [16].

Therefore, we are interested in the singular value decomposition of matrices with low correlation in both the descending space and the column space. Another factor need to consider is the nature of the sampling. Too special a sampling method can also lead to the failure of matrix filling. Therefore, we are concerned with the problem of matrix filling with nearly uniform distribution of data sampling, as shown in Fig. 1.

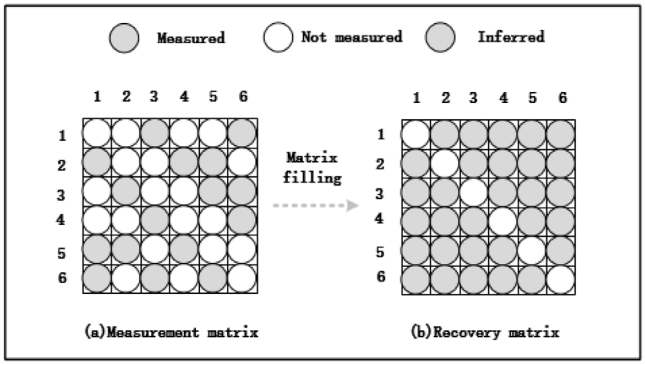


Fig. 1. Schematic diagram of matrix filling.

4.3. Performance evaluation index

The performance of wireless sensor network self — positioning system and algorithm directly affects its availability. How to evaluate is an issue that needs to be discussed in depth. The following are some common evaluation criteria.

(1) Positioning accuracy. The primary evaluation index of the positioning algorithm is the positioning accuracy, which can be expressed by the positioning error or the proportion between the positioning error and the wireless communication distance of the node [17].

(2) Node size. Different positioning systems or algorithms may also be able to locate within the campus, within the building, within a one-story building, or just within a room.

In addition, given a certain amount of infrastructure or a period of time, how much a technology can target is also an important evaluation indicator [9].

(3) Base station node (anchor node) density. Anchor node positioning usually relies on manual deployment or satellite positioning. The way of manually deploying anchor nodes is not only restricted by the network deployment environment, but also severely restricted the scalability of the network and the application. With satellite positioning, the usage rate of anchor nodes will increase greatly, thus increasing the cost. Therefore, anchor node density is also one of the important indexes to evaluate the performance of positioning system and algorithm [11].

(4) Node density. In a wireless sensor network, the increase of node density not only means the increase of network deployment cost, but also leads to limited bandwidth congestion due to communication conflict between nodes. The accuracy of many localization algorithms is affected by node density. Node density is usually expressed as the average connectivity of the network.

(5) Fault tolerance and adaptability. Generally, the positioning system and algorithm need ideal wireless communication environment and reliable network node equipment [10]. However, many problems are often encountered in real applications, such as multipath interference, attenuation, etc. Therefore, the software and hardware of the positioning system and algorithm must have strong fault tolerance and self-adaptability, and can correct errors through automatic adjustment or reconstruction, adapt to the environment, reduce the impact of various errors, so as to improve the positioning accuracy.

(6) Power consumption. Power consumption is one of the most important factors affecting the design and implementation of localization algorithm. Because the battery energy of sensor node is limited,

the calculation amount, communication cost, storage cost and time complexity that are closely related to the power consumption are a set of key indicators on the premise of ensuring the positioning accuracy.

(7) Cost. The cost of a positioning system or algorithm can be evaluated in several different ways. Time cost includes the installation time, configuration time, and positioning time of a system. Space cost includes the infrastructure required for a positioning system or algorithm and the number of network nodes, hardware size, etc. Capital cost includes the infrastructure needed to implement a positioning system or algorithm and the total cost of node equipment.

The above seven performance indexes are not only the criteria for evaluating the positioning system and algorithm of wireless sensor network, but also the optimization criteria for its design and implementation. However, these performance metrics are interrelated and must be weighed against the specific requirements of the application in order to select and design appropriate positioning technologies that can be applied to current conditions.

5. Augmented Lagrange matrix reconstruction and localization algorithm

5.1. Extrapolation based on augmented Lagrange measurements

Given a sensor network, the purpose of this paper is to estimate the distance information of other node pairs through matrix filling in the case that the distance information of a few node pairs is known, and then determine the node position based on this information combined with MDS. The key problem here is to reconstruct the distance between all nodes and fill in the distance matrix.

The size of the neighborhood directly determines the topological stability, robustness and computational efficiency of the algorithm. If the neighborhood is too large, the data structure will be broken and the "short-circuit" edge will be generated, so that the neighborhood cannot correctly express the data set structure, and the neighborhood too small will affect the continuity of the manifold structure [18]. Traditional algorithms determine the neighborhood by setting neighborhood parameters in advance and judge the selection of parameters by the size of mapping "quality" (residual matrix). By finding and removing the "short circuit" edge in the neighborhood graph, the algorithm is no longer sensitive to the size of the neighborhood, so as to avoid the problem that it is difficult to effectively select the size of the neighborhood and improve the operation efficiency. Spatial outlier detection algorithms are based on clustering, distance, density and statistics, etc [19]. When spatial data has serious self-correlation and heterogeneity and other constraints, the statistical algorithm has a good effect. PLS [20] method is a suitable for regression and classification research of the second generation of modeling methods, widely used in machine learning and chemical analysis etc., use covariance between the input and output vector potential information extraction feature, at the same time realize the multiple linear regression, principal component analysis and canonical correlation analysis, the less sample size requirements, running speed, easy to distinguish between information and noise effectively, it is suitable for the situation that the independent variable has serious multiple correlation. The contribution graph of sample points USES the contribution of each variable in the sample point to the latent variable in the explanatory variable space to analyze its influence on the general trend, and can detect outliers that have a great influence on the model. As a unique outlier detection technology of PLS method, it has a strong detection capability.

PLS method adopts the non-parametric estimation method [20], which avoids the LS method's strict parameter requirements and has no distribution requirements on the sample data, and can obtain the asymptotically correct estimation. The algorithm combines the advantages of PLS and nuclear technology to realize the extension of PLS linear regression to high-dimensional nonlinear space. Meanwhile, the input matrix in the regression model is the centroid matrix, which eliminates the influence of bias terms and reduces the dimension and computational complexity of regression coefficient matrix.

The distance square matrix is a low-rank matrix without considering noise. As the measurement method is affected by the environment and the accuracy of the measuring equipment, the measurement data has noise. Sparse signal reconstruction is an important part of compression measurement theory [21]. The compression measure is E. Candes and D. Donohoet al. proposed a new signal sampling measurement method in 2004, The main idea is that the signal with sparse characteristics can be sampled at a lower sampling frequency than Shannon, and the original signal can be recovered with high precision through numerical optimization [22]. Its theoretical basis is signal sparsity, measurement matrix, sparse convex optimization and random matrix principle [23]. The principle of compression measurement has been widely used in image processing, wireless communication and magnetic resonance imaging. Because there may be noise pollution in the scene, the traditional modeling method cannot meet the modeling requirements. By decomposing the sparse and low rank matrix, we can find the low dimension feature space from the high dimension observation samples polluted by strong noise or partially missing, effectively recover the low dimension subspace of the observation samples, and recover the damaged observation samples. Therefore, in this paper, the measured distance square matrix A is modeled as the sum of α low-rank matrix D and α sparse noise matrix Z, that is, $A = D + Z$. A is the measured distance, Z is the measurement error caused by noise. According to the definition of matrix filling problem, solving the complete distance square matrix from a few measurement data can be transformed into the following optimization problem:

$$\min \|D\| + \gamma \|P_{\Omega}(Z)\|_F^2$$
$$\text{Subject to } P_{\Omega}(D + Z) = P_{\Omega}(A) \tag{5}$$

Its augmented Lagrangian function is:

$$L(D, Z, Y, \mu) = \|D\| + \gamma \|P_{\Omega}(Z)\|_F^2 + \langle Y, A - D - Z \rangle + \frac{\mu}{2} \|A - D - Z\|_F^2 \tag{6}$$

In order to solve Eq. (6), this paper designs a matrix filling algorithm based on augmented Lagrange (ALM).

5.2. MDS–ALM positioning

Multidimensional scaling analysis from psychology research techniques is used in the wireless sensor network positioning in recent years [9]. The multidimensional scaling analysis method mainly uses the anisotropy information of each node, such as the distance between nodes and the interconnection of nodes, to establish the objective function and optimize the objective function, and finally obtain the position information of each node.

The algorithm is given by nonlinear mapping space linear inseparable problem within the corresponding high dimensional feature space is converted to linear separable problem, but there are nonlinear mapping form choice, and the problem of the feature space dimension higher dimension would produce operation "dimension disaster, according to the theory of Hilbert space (Hilbert space under the sufficient and necessary condition of positive definite kernel function and determine) [9], K need to satisfy the Mercer condition, for the positive semi-definite matrix, because of the influence of noise and geodesic distance approximation, there is no guarantee that it is qualitative, the algorithm generalization ability is poor.

Nuclear technology uses kernel function to replace inner product operation in nonlinear mapping to avoid mapping form selection and dimension disaster. This algorithm is regarded as the kernel principal component analysis method. It adopts the method of increasing constant to transform K into Mercer kernel matrix, which can simultaneously guarantee the invariance of geodetic distance and semi-positive character. It USES the diagonalization of kernel matrix to obtain the low-dimensional embedding of high-dimensional data, so as to improve

the generalization ability of the algorithm. PLS method adopts the non-parametric estimation method, which avoids the LS method's strict parameter requirements and has no distribution requirements on the sample data, and can obtain the asymptotically correct estimation. The algorithm combines the advantages of PLS and nuclear technology to realize the extension of PLS linear regression to high-dimensional nonlinear space. Meanwhile, the input matrix in the regression model is the centroid matrix, which eliminates the influence of bias terms and reduces the dimension and computational complexity of regression coefficient matrix. The following is a brief description of its calculation process and meaning in the positioning problem.

As shown below, S represents the symmetric matrix composed of the distance between nodes, and the n-dimensional square distance matrix can be expressed as D:

$$S=\begin{bmatrix} 0 & d_{12} & d_{13} & \cdots & d_{1n} \\ d_{21} & 0 & d_{23} & \cdots & d_{2n} \\ d_{31} & d_{32} & 0 & \cdots & d_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ d_{n1} & d_{n2} & d_{n3} & \cdots & 0 \end{bmatrix} D=\begin{bmatrix} 0 & d_{12}^2 & d_{13}^2 & \cdots & d_{1n}^2 \\ d_{21}^2 & 0 & d_{23}^2 & \cdots & d_{2n}^2 \\ d_{31}^2 & d_{32}^2 & 0 & \cdots & d_{3n}^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ d_{n1}^2 & d_{n2}^2 & d_{n3}^2 & \cdots & 0 \end{bmatrix} \tag{7}$$

The bicentric representation of matrix B is $B=(-1/2)L_nDL_n$

By eigenvalue decomposition of covariance matrix D and normalization of the corresponding eigenvector matrix [24], the relative position coordinates of sample points in the feature space of are obtained. It is assumed that the distribution of sample points in the manifold structure is consistent, that is, all sample points contribute the same to the latent variables of the data set.

If the coordinate matrix of all sensor nodes is known, the matrix B can also be expressed as

$$B=X'X'^{T} \tag{8}$$

As matrix B can be obtained from matrix D and centralization matrix L, and as a result, only the vector needs to be solved under the condition of known B, that is the coordinates of nodes. In MDS algorithm, use singular value decomposition method, that is, carry out singular value decomposition on B to obtain the corresponding coordinate solution.

Consider a set of sensor nodes V working on a two-dimensional plane. The nodes are deployed in a random and uniform manner and are dense enough; The maximum communication radius between nodes is r. Based on ALM and MDS, design MDS–ALM localization algorithm.

Algorithm step 1: The shortest path between each node whose radius is less than equal r is used to form the sampling matrix A, and the noise filtering matrix is filled to obtain the distance between each node, and obtain the distance matrix $D=\left[d_{ij}^2\right]_{n*n}$;

Algorithm steps 2, 3, 4: to: D matrix using the classical multidimensional scaling (MDS) technology to obtain the relative coordinates $R=\left[r_{ij}\right]_{2*n}=\left[R_1,R_2,\ldots,R_n\right]$;

Algorithm step 5: Given enough anchor nodes, the relative position is converted to absolute position.

5.3. Simulation experiments and results

5.3.1. Experimental analysis

In order to make the generated topology clear, only 46 common sensor nodes and 4 anchor nodes are randomly deployed in the 100 m × 100 m square area. As shown in Fig. 2, pentagonal star represents anchor node, square represents common sensor node, and edge represents that the node is connected with neighboring nodes. The wireless sensor network is arranged on the 20 m × 20 m square, and the square is divided into small grids with a size of 1 m by 1 m, so the total grid number N = 400.40 sensor nodes were randomly arranged in 400 grid points.

The sensor node processor and communication unit adopt the CC2430 wireless communication chip produced by TI company, and the chip has built-in microcontroller unit, DMA processing, 128 KB and 8 KB random access memory, and has 8-way 14-bit mode/number conversion function, working frequency band 2.4 GHz, ZigBee protocol is adopted as the communication protocol, the communication range is 50-70 m, multi-hop communication can be achieved between nodes. Audio sensing module adopts Panasonic WM64PC omnidirectional audio sensor with sensitivity of (−45±4) dB and impedance less than 2.2 k Ω, frequency response range of 20 Hz ~ 16 kHz, 58 dB, signal-to-noise ratio is less than 4 kHz sampling frequency. In the center of the square, a laptop is arranged as the center node, and the main frequency is 1.40 GHz AMD Atholon XP1600+ processor and 1 GB DDR2 memory.

The sensor node calculates the sound signal intensity once per second, i.e., sampling point $L = 4000$. The sound source target is simulated with a sensor node equipped with a buzzer. The sensor node measures the received sound signal energy, sends the measured data, its time stamp and node ID to the center node through multi-hop communication, and the center node runs the target positioning algorithm to realize multi-target positioning. In the experiment, firstly, the amplitude of the sound source target signal and the gain of the sensor node are calibrated, and then change the sound source target position to carry out the sound source target positioning experiment.

In the experiment, firstly, compare sparse signal reconstruction target location (SSR-GMP) based on greedy matching tracking, sparse signal reconstruction target location (SSR-OMP) based on orthogonal matching tracking, objective calibration bit reconstruction based on L1 norm minimization (SSR-L1) and target location estimation based on simplex search (MLE). Then analyze the influence of system parameters such as SNR, number of nodes M and number of targets K on the performance of target positioning. Target positioning performance is represented by average target positioning error and failure rate. Definition: the positioning fails when the distance between the estimated position and the real position is more than 3 m.

The target location failure rate is expressed by the ratio of the number of target location failures to the total number of targets. The lower the target location failure rate, the more stable the algorithm performance. When the target positioning fails, the estimated position of the target may be very far from the actual position of the target, so when calculating the average positioning error of the target, the positioning failure node error is replaced by 3 m.

In Fig. 2, the communication radius between nodes is R, and the average connectivity of the network is C. In the experiment, it is assumed that the connection and measurement information of the locating and calculating points are known, and the selected anchor nodes are randomly deployed and not on a straight line. The distance between nodes within the communication radius is measurable. Considering the actual measured noise factors, Gaussian noise was added to a small part of the distance data in the experiment to replace the impact of noise in the actual environment.

The average positioning error widely used in previous positioning algorithms is used to measure the performance of the positioning algorithm, which is expressed as

$$error=\frac{\sum_{i=m+1}^{n}\left\|T_{est}^{i}-T_{real}^{i}\right\|^2}{(n-m)R}\times 100\% \tag{9}$$

Among them n is the total number of nodes, m is the number of anchor nodes, and R is the wireless communication radius of nodes.

5.3.2. Simulation results

According to the algorithm steps, the MDS-MAP POSITIONING algorithm and OPTSPACE POSITIONING algorithm were simulated. Fig. 3 respectively represent the actual network topology generated when the communication radius $R = 28$ m and connectivity $C = 9.84$, the real location network topology generated by the mds-alm algorithm and the real location network topology generated by the OPTP algorithm.

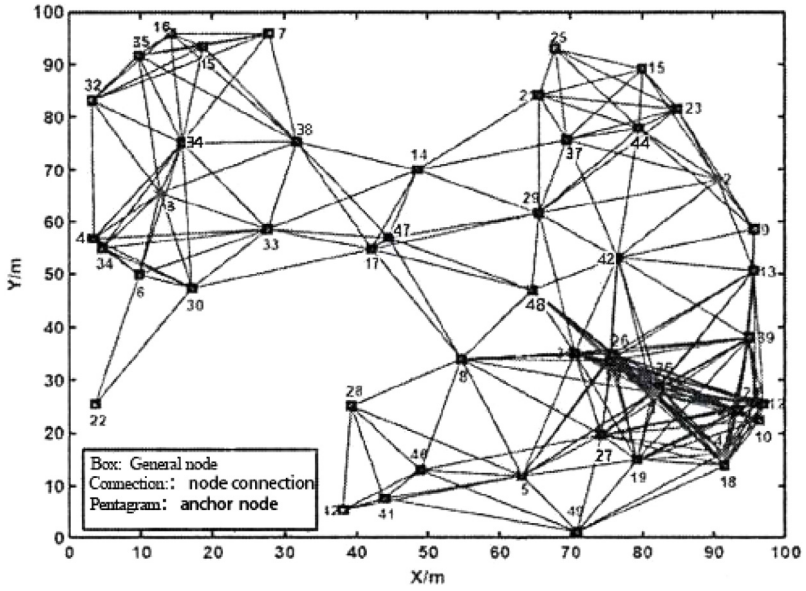
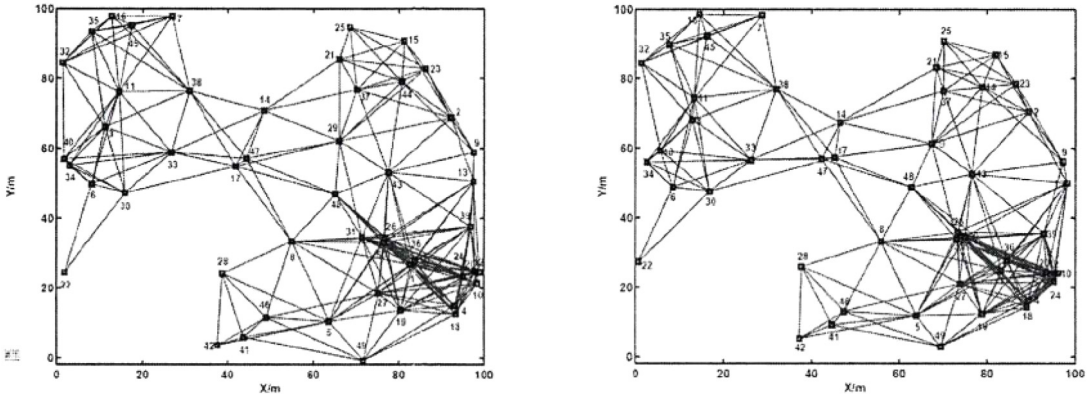


Fig. 2. Actual network topology.



(a)MDS-ALM (b)OPTASPACE POSITIONING

Fig. 3. The location network topology.

Comparing Figs. 3(a) and 3(b), it can be seen that the real location network topology obtained by mds-alm is closer to the real network topology generated by OPTSPACE POSITIONING, and the topology generated by mds-alm is more consistent with the real topology.

When the noise level changes, comparing the positioning errors of the classical MDS and mds-alm, it can be seen that the positioning errors of both algorithms increase with the increase of the noise level, and the mds-alm algorithm shows a slow trend of increasing the error compared with MDS. Under the same noise condition, MDS–ALM algorithm reduces the influence of noise by eliminating the specific points in the sample set. At the same time, PLS method is adopted to improve the accuracy of solution, with minimum error and relatively stable change. When the sample increases, the MDS algorithm needs to recalculate the whole new sample set including the original sample set, while the mds-alm algorithm only needs to calculate the geodesic distance between the newly added sample and the original sample, which reduces the computational complexity, the algorithm execution speed is fast. When the number of samples is greater than 350, the execution time of both algorithms increases sharply, because MDS and mds-alm transform the positioning problem into SDP problem, and SDP has a high algorithm complexity. When the number of samples is large, the execution time of SDP algorithm is long.

As shown in Fig. 3, the comparison between the two positioning algorithms is as follows: square represents the node coordinate after positioning by the algorithm, cross represents the node coordinate after positioning by the algorithm, and asterisk represents the absolute coordinate of anchor node. The shorter the line, the higher the accuracy of the positioning algorithm; otherwise, it was lower. Comparing the length of each line, it can be found that the positioning accuracy of MDS–ALM is obviously better than OPTP.

6. Conclusion

Based on distance is only part of the incomplete information and carry the noise data of the wireless sensor network, this paper proposes a localization algorithm based on matrix filling MDS–ALM, is very suitable for cannot get enough of measuring distance and vulnerable to the external environment pollution situation, for the wireless sensor network node location provides a new solution. Theoretical and experimental analysis shows that the algorithm performs well in predicting unmeasured distance data and separating noise and anomaly, but there is room for further improvement. In the next step, the noise and anomaly of the model will be further modeled, and the measured distance square matrix will be modeled into low-rank distance square matrix, sparse noise matrix, and anomaly matrix, for finer separation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Ping Wang: Conceptualization, Writing - review & editing. **Gang Tu:** Data curation, Resources, Software.

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